

A Box Plot with ANOVA analysis depicts how a categorical variable (e.g. Disease) differs across a continuous variable (e.g. Age).

Step 1: Select samples

Drag and drop terms of interest into the Subset boxes on the **Comparison** tab. Include all terms to be used in the analysis.

Click on the **Summary Statistics** tab to review the number of samples that will be included in the analysis. Confirm that the set of samples available is sufficient for running a meaningful analysis.

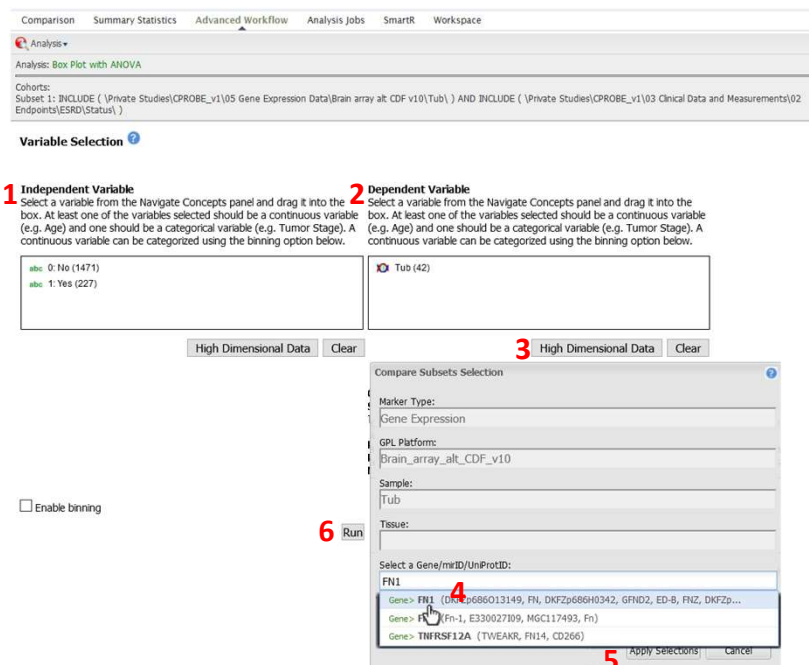
Step 2: Set up analysis

Click on the **Advanced Workflow** tab and select “Box Plot with ANOVA” from the Analysis dropdown list.

Drag terms from the Navigate Concepts tree into the Independent Variable (1) and Dependent Variable (2) boxes. You can add multiple terms or drag over a folder to add all the terms within it. One box should contain a continuous variable (e.g. expression) and the other box should contain a categorical variable (e.g. ESRD Status).

To use High Dimensional Data, drag a high dimensional data node into the variable box, then click on the “High Dimensional Data” button (3). In the pop-up box, search for a HUGO gene symbol, select the desired gene from the drop down (4), and then hit “Apply Selections” (5).

Click the “Run” button (6). A popup box will show a progress bar as the data is compiled and analyzed by tranSMART. Some analyses, especially those with high dimensional data, may take several minutes to run.



The screenshot shows the 'Comparison' tab in the tranSMART interface. The 'Analysis' dropdown is set to 'Box Plot with ANOVA'. The 'Cohorts' section shows the selected subset: 'Subset 1: INCLUDE (\Private Studies\CPROBE_v1\05 Gene Expression Data\Brain array alt_CDF_v10\Tub\) AND INCLUDE (\Private Studies\CPROBE_v1\03 Clinical Data and Measurements\02 Endpoints\ESRD\Status\)'. The 'Variable Selection' section has two boxes: 'Independent Variable' (1) containing 'age' and 'sex', and 'Dependent Variable' (2) containing 'Tub (42)'. The 'High Dimensional Data' button (3) is visible. The 'Compare Subsets Selection' dialog box is open, showing the 'Marker Type' as 'Gene Expression', 'GPL Platform' as 'Brain_array_alt_CDF_v10', 'Sample' as 'Tub', and 'Tissue' as 'Tub'. The 'Select a Gene/miID/UnProtID:' field is populated with 'FN1'. The 'Run' button (6) is highlighted. The 'Apply Selections' button (5) is also visible.

Step 3: Review results

Results will appear underneath the variable selection boxes. The Independent variable will appear along the X-axis and the Dependent variable will be the Y-axis (7). For High Dimensional Data, the selected gene will be noted as “Pathway” above the chart, and the reporter from the platform that matches the gene will be visible next to and below the visualization.

Scroll below the box plot to see ANOVA results (8).

